

Elementary Problems Diophantine Equation

Version 1.0

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Basic Work

1. (IMO shortlist 1984) Prove:

(a) There are infinitely many triples of positive integers m, n, p such that

$$4mn - m - n = p^2 - 1.$$

2. Find all integers m and n such that $mn + 2m = n + 3$.
3. Find all integers m and n such that $2mn + 3m = 3n + 5$.
4. (Spain MO 1995 Problem 4) Given a prime number p , find all integer solutions of $p(x + y) = xy$.
5. Find all integers m and n such that $107m + 43n = 4$.
6. Find all positive integers m and n such that $4m^2 = n^2 + 35$.
7. Find all positive integers n such that $n^2 + 12n + 8$ is a perfect square.
8. (IMO shortlist 1966) For which real numbers p does the equation $x^2 + px + 3p = 0$ have integer solutions?
(Remark: One may assume both solutions are integers.)
9. Let m and n be positive integers such that $[m, n] + (m, n) = m + n$. Prove that one of m, n divides the other one.
10. Find all positive integers m and n such that $m^2n + n^2 + m = mn^2 + m^2 + n + 30$.
11. Find all prime numbers p such that both $p - 1$ and $2p - 1$ are perfect squares.
12. (Flanders MO 2005 Problem 4) If n is an integer, then find all values of n for which $\sqrt{n} + \sqrt{n + 2005}$ is an integer as well.
13. Prove that for any positive integer n , there exist n positive integers (not necessarily distinct) whose sum is equal to their product.
14. (IMO shortlist 1969) Let a, b, x, y be positive integers such that a and b have no common divisor greater than 1. Prove that the largest number not expressible in the form $ax + by$ is $ab - a - b$. If $N(k)$ is the largest number not expressible in the form $ax + by$ in only k ways, find $N(k)$.
(Remark: Indeed, x, y should be nonnegative integers. Otherwise, the conclusion does not hold. Also, ‘only’ k ways should be replaced by ‘at least’ k ways.)
15. Find all positive integers m and n such that $m^2 - 3n^2 = 1$.

16. Find all integers m and n such that $m^2(n-1) + n^2(m-1) = 1$.
17. (USAMO 1987 Problem 1) Determine all solutions in nonzero integers a and b of the equation
- $$(a^2 + b)(a + b^2) = (a - b)^3.$$
18. (CWMO 2007 Problem 6) Find all positive integers n such that there exist nonzero integers $x_1, x_2, \dots, x_n, y$ satisfying
- $$\begin{cases} x_1 + \dots + x_n = 0, \\ x_1^2 + \dots + x_n^2 = ny^2. \end{cases}$$
19. Prove or disprove the following.
- (a) The equation $12345m + 13579n = 270$ has infinitely many integer solutions.
 - (b) The equation $4m^2 - n^2 = 11111$ has finitely many integer solutions.
 - (c) The equation $3m^2 - 2n^2 = 1$ has finitely many integer solutions.
 - (d) Let $f(x)$ and $g(x)$ be polynomials with integer coefficients. Let n be an integer. Then $m^2 + f(n)m + g(n) = 0$ has an integer solution in m if and only if $f(n)^2 - 4g(n)$ is a perfect square.
 - (e) If a Diophantine equation has a solution modulo p for infinitely many prime numbers p , then it has integer solutions.

Congruence

20. Find all integers m and n such that $12m + 27n = 100$.
21. Let k be a positive integer. Prove that there do not exist odd integers $a_1, a_2, \dots, a_{2k}$ such that
- $$\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_{2k}} = 1.$$
22. (IMO shortlist 1969) Which natural numbers can be expressed as the difference of squares of two integers?
23. Find all positive integers m and n such that $m! + 76 = n^2$.
24. (All-Soviet Union MO 1974) Find the smallest positive integer which can be represented in the form $36^m - 5^n$.
- (Remark: We may assume m, n are positive integers.)
25. (Greece MO 2001 Problem 2) Prove that there are no positive integers a, b such that $(15a + b)(a + 15b)$ is a power of 3.

26. Find all prime numbers $p < 50$ such that $3^n + p$ is a perfect square for some positive integer n .
27. (IMO 1984 Problem 6) Let a, b, c and d be odd integers such that $0 < a < b < c < d$ and $ad = bc$. Prove that if $a + d = 2^k$ and $b + c = 2^m$ for some integers k and m , then $a = 1$.

Size

28. (IMO shortlist 1966) Does there exist an integer z that can be written in two different ways as $z = x! + y!$, where x, y are natural numbers with $x \leq y$?
29. For each positive integer n , the sum of digits of n is denoted by $S(n)$. Find all positive integers n such that $S(n^2) = n$.
30. (IMO 1968 Problem 2) Find all natural numbers x such that the product of their digits (in decimal notation) is equal to $x^2 - 10x - 22$.
31. (Brazil MO 1983 Problem 1) Show that there are only finitely many solutions to

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{1983}$$

in positive integers.

32. Find all positive integers a, b, c such that

$$\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)\left(1 + \frac{1}{c}\right) = 2.$$

33. Find all positive integers a, b, c such that $a + b + c + ab + bc + ca = abc + 1$.
34. (All-Russian MO 1966) Can both $x^2 + y$ and $x + y^2$ be squares for natural numbers x, y ?
35. Find all positive integers n such that $n^2 + 12n + 8$ is a perfect square.
36. (CWMO 2002 Problem 1) Find all positive integers n such that

$$n^4 - 4n^3 + 22n^2 - 36n + 18$$

is a perfect square.

37. (Baltic Way 2011) Determine all pairs (p, q) of primes for which both $p^2 + q^3$ and $q^2 + p^3$ are perfect squares.
38. (Iran MO 2nd Round 2008 Problem 4) $\mathbb{N}$ is the set of positive integers and $a \in \mathbb{N}$. We know that for every $n \in \mathbb{N}$, $4(a^n + 1)$ is a perfect cube. Prove that $a = 1$.

Greatest Common Divisor

- 39. (All-Russian MO 1964) Prove that for any natural numbers m, k, n with $n > 1$, we cannot have $m(m+1) = k^n$.
- 40. Find all positive integers m and n such that $m(m^2 + 8m + 1) = n^2$.
- 41. Find all nonnegative integers m and n such that $m^4 - 2n^2 = 1$.
- 42. Find all nonnegative integers m and n such that $mn^2 = m^2 + n$.
- 43. Find all positive integers m and n such that $m^2 = 2^n + 49$.
- 44. (IMO shortlist 1979) Find all natural numbers n for which $2^8 + 2^{11} + 2^n$ is a perfect square.
- 45. Prove that the integer $33 \cdots 3$ cannot be a perfect power.

Record

v1.0: problems 1–45